

Please write clearly in block capitals.

Centre number

Candidate number

Surname

Forename(s)

A-level

ENVIRONMENTAL SCIENCE**Paper 2 (Generated practice paper - Set 28)**

Assessed: The living environment, Biological resources, Sustainability, Research methods

Time allowed: 3 hours**Materials**

For this paper you may use:

- a calculator.

Instructions

- Use black ink or black ball-point pen. Pencil should only be used for drawing.
- Fill in the boxes at the top of this page.
- Answer **all** questions 1 to 10 and **one** essay from question 11.
- You must answer the questions in the spaces provided. Do not write outside the box around each page or on blank pages.
- Do all rough work in this book. Cross through any work you do not want to be marked.

Information

- The marks for questions are shown in brackets.
- The maximum mark for this paper is 120.
- All questions should be answered in continuous prose.
- You will be assessed on your ability to:
 - use good English
 - organise information clearly
 - use specialist vocabulary where appropriate.

For Examiner's Use

Question	Mark
1	
2	
3	
4	
5	
6	
7	
8	
9	
10	
11	
TOTAL	

Answer all questions in the spaces provided.

0 1 . 1

Describe **two** case studies of biological control, one successful and one that caused problems.

[4 marks]

.....

.....

.....

.....

.....

.....

.....

.....

.....

0 1 . 2

Explain how pheromone traps can be used in pest control.

[2 marks]

.....

.....

.....

0 1 . 3

Explain why the use of antibiotics as growth promoters in livestock is a concern for human health.

[3 marks]

.....

.....

.....

.....

.....

.....

.....

0 1 . 4

Give **one** way farmers can support crop pollinators.

[1 mark]

.....

10

0 2 . 1

Compare selective breeding, asexual reproduction and cross-breeding as methods of improving livestock or crops.

[6 marks]

0 2 . 2

Explain how artificial insemination and embryo transfer increase the rate at which desirable characteristics spread through a livestock population.

[4 marks]

10

03 . 1

Explain why introduced species often have a greater effect on island communities than on mainland communities.

[2 marks]

03 . 2

For each of the following, describe an example of an introduced species and explain its effect on an indigenous species.

[6 marks]

Introduced competitor

.....

.....

.....

Introduced predator

.....

.....

.....

Introduced pathogen

.....

.....

.....

03 . 3

Explain how hybridisation with an introduced species threatens the red deer and the Scottish wildcat.

[2 marks]

.....

.....

.....

10

0 4 The Rio Bravo Conservation and Management Area in Belize exploits mahogany under a sustainable management plan while conserving a rainforest of over 200 tree, 400 bird and 70 mammal species.

0 4 . **1** Describe **five** features of the timber harvesting at Rio Bravo that make it sustainable.

[5 marks]

1

2

3

4

5

0 4 . **2** Explain why involving local communities is essential to the success of the reserve.

[3 marks]

.....

.....

.....

.....

.....

.....

.....

0 4 . **3** Explain why selective logging is better for wildlife than clear-felling, and give **one** disadvantage.

[2 marks]

10

0 5

Table 1 shows life-history features of three fish species and the age at which they reach a catchable size.

Table 1

Species	Age at catchable size / years	Age at first breeding / years	Eggs per female per year	Maximum lifespan / years
Atlantic cod	2	4	Millions	25
Orange roughy	20	30	Thousands	150
Herring	2	2	Tens of thousands	20

0 5

. 1

Use Table 1 to explain why orange roughy populations collapsed within ten years of large-scale fishing beginning in the late 1970s.

[4 marks]

0 5

. 2

Explain why herring populations recover from overfishing much faster than cod.

[2 marks]

0 5 . 3

Explain why the total mass of fish landed can stay the same while a population is being overfished.

[2 marks]

0 5 . 4

Describe how the Greenland shark and blue (common) skate illustrate the threat of by-catch to k-selected species.

[2 marks]

10

0 6 . 1

Explain **four** properties of water that make it essential for life.

[4 marks]

0 6 . 2

Describe how photosynthetic organisms changed the atmosphere and explain how this allowed life to colonise the land.

[4 marks]

06 . 3

Explain how carbon sequestration by early photoautotrophs helped maintain a suitable temperature on Earth.

[2 marks]

10

07

Figure 1 shows a simplified Antarctic food web.

Figure 1

Trophic level	Organisms
Producers	Phytoplankton, ice algae
Primary consumers	Zooplankton, krill
Secondary consumers	Small fish, squid, baleen whales, crabeater seals, penguins
Higher consumers	Toothfish, albatross, leopard seal, killer whale

07 . 1

Explain why the sea around Antarctica is so productive despite the cold.

[3 marks]

07 . 2

Use Figure 1 to explain why fishing for krill and for Patagonian toothfish threatens other species.

[3 marks]

07 . 3

Explain the importance of Antarctica in controlling global natural cycles.

[3 marks]

07 . 4

Explain why the area of winter sea ice around Antarctica has increased in recent years even though air and sea are warming.

[1 mark]

10

08 . 1

Describe how agriculture alters the hydrological cycle.

[3 marks]

.....

.....

.....

.....

.....

.....

08 . 2

Explain how each of the following consumer choices can reduce the environmental impact of food production.

[4 marks]

Seasonal / local food

.....

.....

Organic food

.....

.....

08 . 3

Explain why post-war European governments introduced grants and guaranteed prices, and describe **one** environmental consequence.

[3 marks]

.....

.....

.....

.....

.....

.....

10

09

A student sampled the ground-flora of two woodlands. Table 1 shows the results for woodland A.

Table 1

Species	Number of individuals (n)
Bluebell	40
Wood anemone	30
Dog's mercury	20
Bramble	10
Total (N)	100

09

1

Use the formula $D = N(N - 1) / \sum n(n - 1)$ to calculate Simpson's index of diversity for woodland A.

[3 marks]

Answer

09

2

Woodland B had an index of 1.6. State what this shows and give **one** limitation of using the index alone to compare the woodlands.

[2 marks]

5

Discuss the reasons why the conservation of biodiversity is important for human society.

ESP/7447/2/SET28

Extra space

10 . 2 State what is meant by **biodiversity**.

[1 mark]

10

Write an essay on one of the following topics.

$$\boxed{1} \boxed{1} . \boxed{1}$$

Discuss how ecological monitoring and the development of new technologies have improved the planning and evaluation of conservation strategies.

[25 marks]

OR

1 1 . 2

Natural systems have remained sustainable over very long periods. Discuss how the features of natural systems could be copied to make human systems more sustainable.

[25 marks]

Indicate which question you have answered by writing its number, **11 . 1** or **11 . 2**, at the top of your answer.

END OF QUESTIONS